

Final examination of Erlang

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1 Merging sorted lists

Question. Write a function `merge/2` which takes two lists of items increasingly sorted by `<` and returns a new sorted list containing the items of both inputs. For instance

```
> final:merge([3,5,8,8,10,12],[1,2,5,6,7,9,11]).  
[1,2,3,5,5,6,7,8,8,9,10,11,12]
```

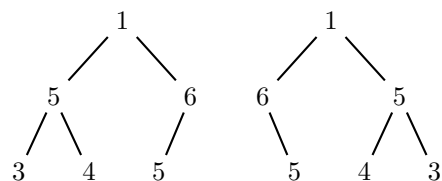
Make two versions: (1) one not tail recursive, (2) one tail recursive.

2 Fun with binary trees

The atom `nil` denotes the empty binary tree and the triple `{Left, Root, Right}` is a non-empty whose root is `Root`, left subtree is `Left` and right subtree is `Right`.

2.1 Mirroring binary trees

Question. Write an Erlang function `mirror/1` which takes a binary tree and returns the same tree as in a mirror. Consider the facing example:



2.2 Summing

Question. Write a function `sum/1` which takes a binary tree whose nodes are integers and returns the sum of the nodes. In the facing example, the sum is 24.

